Signed Additive Forgetting Attention for Generative Recommendation

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Base HSTU code: `facebookresearch/generative-recommenders`; SAFA artifact pending

Abstract

HSTU preserves explicit item pairs, signed SiLU coefficients, pairwise time and position bias, and unnormalized value superposition. We ask whether content-dependent forgetting can improve this mixer without discarding its explicit pairs. *Signed Additive Forgetting Attention* (SAFA) multiplies each HSTU pair coefficient by learned survival across the intervening events. It retains negative relevance and additive evidence while introducing data-dependent intent boundaries. SAFA is deliberately an explicit-pair, quadratic attention rule; "additive" does not denote a complexity claim. On MovieLens-1M, preliminary three-seed SAFA reaches 0.19569 mean NDCG@10 versus 0.19022 for separate raw-HSTU controls. Because those HSTU controls have 416 fewer parameters and ran as separate jobs, this prioritizes SAFA but does not close the claim. We define an exact-inventory matched HSTU–SAFA A/B, with a frozen paired-seed decision rule, as the required test.

Keywords: generative recommendation, HSTU, signed attention, selective forgetting, sequential recommendation

1. Question and Current Evidence

For a causal sequence, suppressing batch indices, write a broad class of mixers as

$$\boldsymbol{z}_i = \mathrm{LN}\left(\sum_{j\le i} a_{ij}\boldsymbol{v}_j\right), \qquad \boldsymbol{y}_i = \boldsymbol{x}_i + W_o(\boldsymbol{u}_i \odot \boldsymbol{z}_i). \tag{1}$$

The question is which properties of a_{ij} matter for recommendation and how forgetting should act on those coefficients.

The research HSTU in this repository [Zhai et al., 2024] uses

$$a_{ij}^{\mathrm{H},h} = \frac{M_{ij}}{N}\,\mathrm{SiLU}\Big((\boldsymbol{q}_i^h)^\top \boldsymbol{k}_j^h + b_{ij}\Big), \tag{2}$$

where h indexes heads, M is the causal mask, and b_{ij} contains learned relative position and time features. This rule is explicit-pair and signed.

The motivating MovieLens-20M runs [Harper and Konstan, 2015] are informative but not a valid proof that Equation (2) beats softmax. The element-wise run `azixyn0z` stopped after epoch 60. Its best NDCG@10 was 0.208284 at epoch 59. The softmax run `dneoj5eq` reached 0.209957 at epoch 100; at the matched epoch 60, element-wise attention was ahead by 0.001804. However, the softmax branch formed one dot product over all Hd_k channels and shared one attention matrix across heads, whereas HSTU formed H independent d_k-dimensional matrices. It also divided the Hd_k dot product by $\sqrt{d_k}$. Thus the run confounds normalization, head structure, and logit scale. Section 4 defines the required rerun.

2. First Principles

2.1. The softmax denominator nearly cancels

Softmax gives

$$a_{ij}^{S,h} = \frac{M_{ij}\exp(s_{ij}^h)}{\sum_{\ell \le i} M_{i\ell}\exp(s_{i\ell}^h)}, \qquad s_{ij}^h = \frac{(\boldsymbol{q}_i^h)^\top \boldsymbol{k}_j^h + b_{ij}}{\tau}. \tag{3}$$

For one head, let $\boldsymbol{r}_i = \sum_{j \le i} \exp(s_{ij})\boldsymbol{v}_j$ and $Z_i = \sum_{j \le i} \exp(s_{ij})$. In the HSTU shell,

$$\mathrm{LN}\left(\frac{\boldsymbol{r}_i}{Z_i}\right) = \mathrm{LN}(\boldsymbol{r}_i) \tag{4}$$

for positive Z_i, exactly when the LayerNorm epsilon is zero and to high accuracy away from a zero-variance aggregate otherwise. With $H > 1$, the code applies one LayerNorm to the concatenation $[\boldsymbol{r}_i^1/Z_i^1; \ldots; \boldsymbol{r}_i^H/Z_i^H]$, so unequal denominators can rebalance heads. They still cannot change the token-mixture direction inside a head. The historical collapsed-head softmax used one shared Z_i, for which Equation (4) applies to the entire concatenated vector. Thus "competition" is at most a cross-head scaling effect in the corrected control, not a complete explanation of the token mixer.

For any pointwise kernel κ, repeating a score-value pair m times gives the pre-normalization term $m\kappa(s)\boldsymbol{v}$. Both exponential and SiLU kernels accumulate multiplicity relative to other value directions. Their meaningful difference is kernel geometry: $\exp(s)$ is positive and grows exponentially, whereas $\mathrm{SiLU}(s)$ is signed, suppresses large negative scores, and is asymptotically linear for positive scores. Score temperature and relative bias interact with this geometry and must be controlled empirically.

The usual $\tau = \sqrt{d_k}$ assumes roughly unit-variance linear query and key coordinates. It is not scale-neutral in this shell. For the MovieLens-20M initialization, $D(0.02)^2 = 0.1024$ before SiLU and $\mathbb{E}[\mathrm{SiLU}(z)^2] \simeq 0.0274$, giving $\mathrm{std}((\boldsymbol{q}^h)^\top \boldsymbol{k}^h) \simeq 0.153$ at $d_k = 32$; division by $\sqrt{32}$ shrinks this to roughly 0.027. Moreover, the historical code uses $(qk + b)/\sqrt{d_k}$, whereas the usual relative-bias convention is $qk/\sqrt{d_k} + b$. These are distinct controls, not implementation details.

This suggests the first hypothesis.

> **H1 (signed pair-kernel geometry).** Recommendation benefits from query-specific, signed, explicit-pair aggregation and the moderate positive tail of SiLU, rather than from removing softmax's row denominator.

2.2. Forgetting while retaining explicit pairs

Old events can become irrelevant when user intent changes. A scalar per-head forget gate $f_t^h \in (0, 1)$ defines the survival of event j at query i,

$$F_{ij}^h = \prod_{\ell=j+1}^{i} f_\ell^h, \qquad F_{ii}^h = 1. \tag{5}$$

This is the bridge used by the Forgetting Transformer (FoX): multiplying an exponential kernel by F_{ij} is equivalent to adding $\log F_{ij}$ to its softmax logit [Lin et al., 2025]. The same survival process can be applied to HSTU without changing its explicit-pair representation.

There is an important asymmetry. Write $\log F_{ij}^h = P_i^h - P_j^h$ for prefix log-survival. In row softmax, P_i^h cancels, so FoSoftmax is exactly $\mathrm{softmax}_j(s_{ij}^h - P_j^h)$: it redistributes mass among keys but cannot change the total coefficient mass, which remains one. That redistribution can still change the value aggregate's norm and direction. With a fixed gate it reduces to a head-specific linear recency bias. Neither cancellation occurs when F_{ij}^h multiplies HSTU's unnormalized signed coefficients.

> **H2 (selective intent boundaries).** The useful temporal inductive bias is data-dependent forgetting across intervening events while retaining HSTU's explicit pairs.

3. Signed Additive Forgetting Attention

We propose Signed Additive Forgetting Attention (SAFA),

$$f_i^h = \sigma((\boldsymbol{w}_f^h)^\top \boldsymbol{k}_i^h + c_h), \tag{6}$$

$$a_{ij}^{\text{SAFA},h} = \frac{M_{ij}}{N} F_{ij}^h \, \text{SiLU}\Big((\boldsymbol{q}_i^h)^\top \boldsymbol{k}_j^h + b_{ij}\Big) . \tag{7}$$

Earlier experiment configs and checkpoints call this same mechanism `FoHSTU`; we retain that string only as a provenance identifier. It has four useful properties. First, HSTU lies in the closure as $f_i^h \to 1$; the matched control sets $F_{ij}^h = 1$ exactly, so the proposal is nested rather than a new block topology. Second, multiplication preserves the sign of every SiLU coefficient. Third, each head can learn a different content-dependent horizon. Fourth, the pairwise position/time bias remains available; the gate adds path information from the events between j and i, which an endpoint-only relative bias cannot represent.

Why signed and additive. Conditioned on the realized gates, SAFA aggregates

$$\boldsymbol{z}_i^h = \sum_{j \le i} a_{ij}^{\text{SAFA},h} \boldsymbol{v}_j^h. \tag{8}$$

There is no row denominator: a second supportive event adds another value contribution instead of taking probability mass from the first. Negative SiLU coefficients can subtract an anti-aligned value direction, while $F_{ij}^h \ge 0$ attenuates magnitude without changing sign. Forgetting can depend on the intervening sequence, but it does not renormalize the surviving keys into competition. These properties motivate the name Signed Additive Forgetting Attention.

Why the gate must be outside SiLU. FoX writes $F \exp(s) = \exp(s + \log F)$. Copying the logit shift into HSTU would give $\text{SiLU}(s + \log F)$, which is not attenuation by F and can change behavior around zero. More generally, a score shift implements multiplicative survival for every s only if a kernel g satisfies

$$g(s + \log F) = F g(s). \tag{9}$$

Under mild regularity, the solutions are exponential kernels. SiLU is not one. Equation (7) is therefore the sign-preserving extension of forgetting to HSTU.

The projected key is used as the gate input so full and cached incremental evaluation share the same state; a separate projection from $\boldsymbol{x}_i$ is a straightforward ablation.

Stable initialization. Initialize $\boldsymbol{w}_f^h = 0$ and choose head time constants T_h logarithmically between $T_{\min}$ and $T_{\max}$. Setting

$$\bar{f}_h = e^{-1/T_h}, \qquad c_h = \log \frac{\bar{f}_h}{1 - \bar{f}_h} \tag{10}$$

makes the initial survival $F_{ij}^h = e^{-(i-j)/T_h}$, after which the gate can become data-dependent. For histories of length 200, the first sweep should use T_h log-spaced from 8 to 256. A timestamp-aware extension replaces one unit step by a normalized elapsed-time step, but it should be tested only after the position-only gate to avoid duplicating the existing time bias.

Exact-inventory matched control. Across L layers, the key-conditioned scalar gate adds exactly $LH(d_k + 1)$ trainable scalars: 416 for the MovieLens-1M model and 4,224 for the MovieLens-20M model. The clean A/B therefore instantiates the same gate parameters in both arms. SAFA uses Equation (6); the matched HSTU arm forces $F_{ij}^h = 1$ and retains a zero-valued dependency on the dormant gate so that parameter names and shapes, optimizer parameter-group schema and initial state, and the DDP used-parameter set are identical. Comparisons with a smaller raw HSTU remain capacity-confounded, even when the difference is numerically small.

SAFA remains $O(N^2 d)$. The current reference implementation explicitly materializes F and therefore adds $O(BHN^2)$ memory. A fused implementation would store only $O(NH)$ gates and prefix sums, compute $\log F_{ij} = P_i^h - P_j^h$ inside each existing score tile, and add one scalar multiplication per pair without writing a second $N \times N$ matrix. The immediate goal is quality and causal identification; kernel optimization follows only if the hypothesis survives.

4. Experiments That Decide the Hypotheses

4.1. Primary exact-inventory matched A/B

The decisive experiment contains only two model behaviors in one clean HSTU implementation. Both instantiate the forget-gate parameters. The matched HSTU arm forces every survival factor to one; the SAFA arm learns Equation (6). All other modules, parameter names and shapes, initialization order, optimizer configuration and initial state, seed, data order, and training steps are identical. The run must fail closed if the trainable inventories differ. Before training, matched HSTU must equal raw HSTU in outputs, loss, every shared-parameter gradient, and one optimizer update; dormant gate gradients must be present and exactly zero. Use paired seeds 42–44 on MovieLens-1M for qualification and repeat the same frozen configuration on MovieLens-20M. The primary statistic is mean NDCG@10 over epochs 96–100; report the paired per-seed delta, sample standard deviation, minimum delta, HR@10, MRR, and equal-work throughput. Before a result is eligible for the paper, the source snapshot and resolved configuration must be checksum-bound, the scheduler array identity and actual QoS must match the frozen submission, and the exported metric record must carry an input checksum.

Before inspecting these clean runs, freeze the qualification rule: the mean paired SAFA-minus-HSTU NDCG@10 delta must be at least 0.002, at least two of three seed deltas must be positive, and no seed may fall below -0.001. A failure rejects this SAFA parameterization rather than triggering a gate sweep on the same seeds.

4.2. Separate normalization control

The historical HSTU–softmax claim requires a same-code-version MovieLens-20M control:

1. HSTU with Equation (2);
2. per-head softmax with logits and values shaped $[B, H, N, N]$ and $[B, N, H, d_v]$, respectively;
3. the same relative bias, causal/padding mask before softmax, projections, block count, dimensions, dropout, seed, data order, and 101 epochs;

4. historical $(qk + b)/\sqrt{d_k}$ scaling, canonical $qk/\sqrt{d_k} + b$, and an unscaled temperature-one control using the exact HSTU score $qk + b$, so temperature and relative-bias scale do not decide the comparison.

Report final epoch, best epoch, matched epoch, matched wall-clock, throughput, and at least three seeds before claiming a normalization advantage. The first seed is a regression run; it is not a publication result.

4.3. Supporting causal controls

These controls interpret the two-arm A/B; they do not create additional model candidates:

Control	Question	Status
Raw tanh HSTU	Does a bounded odd pair activation help without forgetting?	running
No relative bias	How much do pairwise time and position contribute?	completed, one seed
Fixed survival	Is a fixed recency prior sufficient?	completed, one seed
FoSoftmax	Is forgetting useful with an exponential kernel?	completed, one seed
Corrected per-head softmax	Does normalization matter after fixing head shape and scale?	submitted/pending

For the primary A/B, analyze the final checkpoints on deterministic held-out batches for per-head gate and survival quantiles, effective half-life, negative coefficient mass, performance by history length, and elapsed-time gap. These diagnostics interpret the frozen accuracy test; they do not alter it. If SAFA passes the rule while fixed survival and FoSoftmax do not, the supported mechanism is content-dependent survival acting on HSTU's signed, unnormalized coefficients. Otherwise the result does not justify a more complicated gate sweep on the same seeds.

5. Preliminary Results and Limits

Job 1671578 ran full SAFA for three deterministic MovieLens-1M seeds (42–44, 101 epochs) from one checksum-verified source snapshot and common training setup; only full SAFA is shown here. The HSTU row comes from separate deterministic control jobs and is not a within-job comparison. Seed 42 comes from job 1671180 and seeds 43–44 from job 1671258; their HSTU source hashes begin `16f8fcd4` and `399cd3a9`, while the SAFA source hash begins `6e14dd2e`. No cross-snapshot baseline equivalence audit was performed, so the aggregate is descriptive only. The primary statistic is the mean over epochs 96–100:

Variant	Mean NDCG@10	Best NDCG@10	epoch-rate examples/s
HSTU	0.19022 ± 0.00290	0.19235 ± 0.00362	821.1
Full SAFA	0.19569 ± 0.00126	0.19788 ± 0.00165	672.6

This table is not exact-inventory matched: raw HSTU has 313,000 trainable parameters on MovieLens-1M and 38,913,120 on MovieLens-20M, whereas SAFA has 313,416 and 38,917,344. The additions are only 416 (0.133%) and 4,224 (0.011%) parameters, but they are nonzero. The clean exact-inventory matched shell will therefore use 313,416 parameters in both MovieLens-1M arms and 38,917,344 in both MovieLens-20M arms.

NDCG values are mean $\pm$ sample standard deviation. Epoch rates are the arithmetic mean over epochs 1–100 and then over seeds. Full SAFA is 0.00546 above the separate raw-HSTU controls on the primary statistic, but this is descriptive, not the exact-inventory A/B required by Section 4.

Single-seed controls support the mechanism boundary but are not accuracy estimates. Removing relative bias drops best NDCG@10 to 0.165336. Unscaled per-head softmax reaches 0.195046, while fixed and learned FoSoftmax fall to 0.191885/0.190548. In these controls, forgetting improved only when applied to HSTU's signed, unnormalized coefficients; multi-seed confirmation remains required. The corrected, same-snapshot MovieLens-20M softmax comparison is submitted and currently pending; it is not included here.

6. Conclusion

SAFA retains HSTU's explicit pairs, relative time and position, signed SiLU coefficients, and unnormalized additive value aggregation, then adds content-dependent survival between each key and query. Preliminary evidence prioritizes this combination: fixed forgetting and FoSoftmax controls are consistent with neither mechanism alone explaining the observed gain. These diagnostics motivate SAFA but do not establish its advantage. The next primary claim is narrower: reproduce SAFA against the exact-inventory matched HSTU arm on MovieLens-1M and MovieLens-20M before optimizing its explicit-pair kernel.

References

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